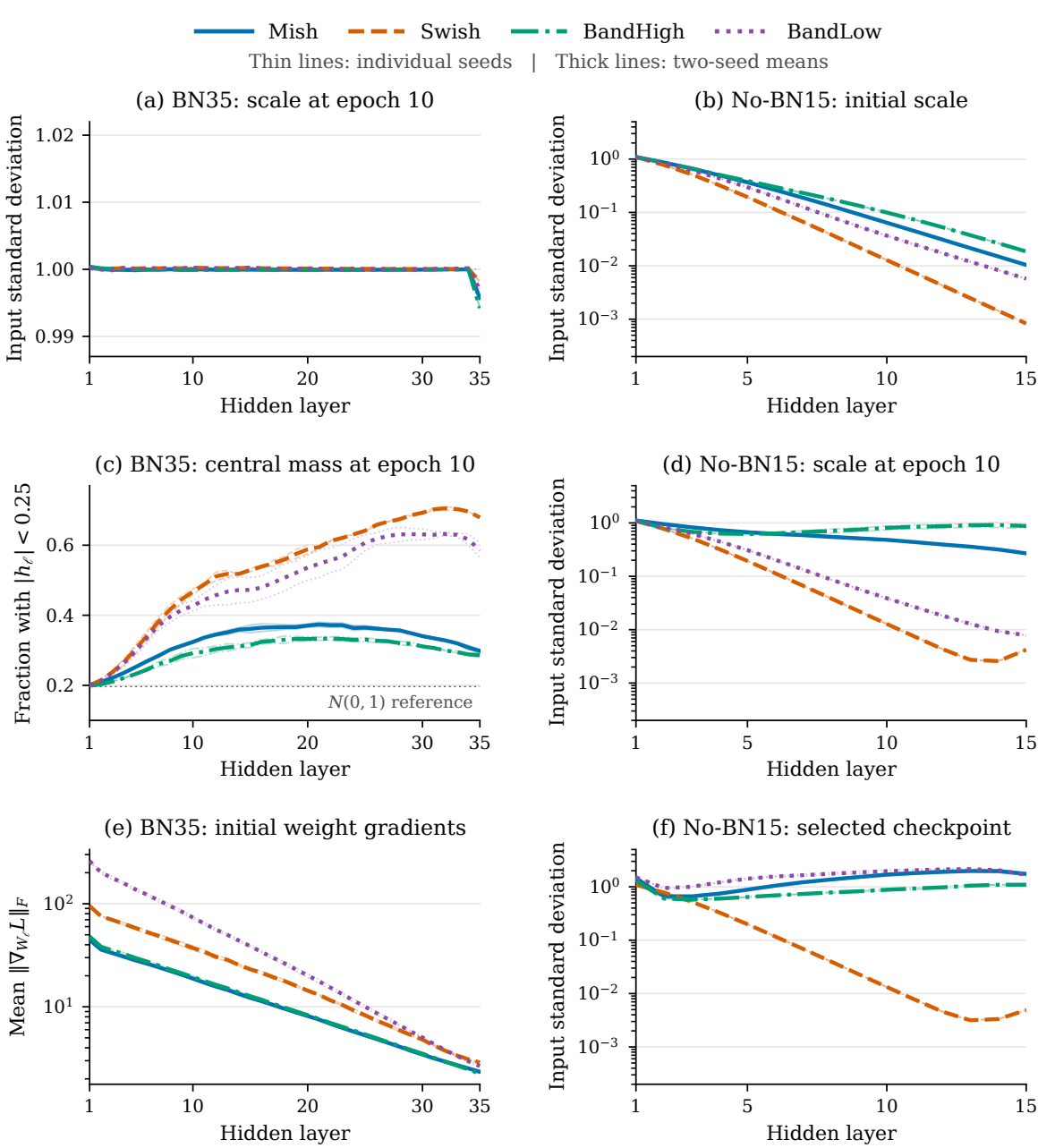
Mish
Swish
BandHigh
BandLow
Thin lines: individual seeds | Thick lines: two-seed means
(a) BN35: scale at epoch 10
Input standard deviation
1.02
1.01
1.00
0.99
1
10
20
30
35
Hidden layer
(b) No-BN15: initial scale
Input standard deviation
10^{0}
10^{-1}
10^{-2}
10^{-3}
1
5
10
15
Hidden layer
(c) BN35: central mass at epoch 10
Fraction with $|h_\ell| < 0.25$
0.6
0.4
0.2
$N(0, 1)$ reference
1
10
20
30
35
Hidden layer
(d) No-BN15: scale at epoch 10
Input standard deviation
10^{0}
10^{-1}
10^{-2}
10^{-3}
1
5
10
15
Hidden layer
(e) BN35: initial weight gradients
Mean $\|\nabla_{W_\ell} L\|_F$
10^{2}
10^{1}
1
10
20
30
35
Hidden layer
(f) No-BN15: selected checkpoint
Input standard deviation
10^{0}
10^{-1}
10^{-2}
10^{-3}
1
5
10
15
Hidden layer